

Python Questions - 01

1. Loan Eligibility Checker (If)

Question:

Write a Python program to check whether a customer is eligible for a loan. If the customer's monthly salary is greater than ₹30,000, display "**Loan Eligible**".

Sample Output:

Enter Monthly Salary: 45000

Loan Eligible

2. Website Login System (If Else)

Question:

Write a Python program to check whether the entered username is admin. If correct, display "**Welcome Admin**", otherwise display "**Access Denied**".

Sample Output:

Enter Username: admin

Welcome Admin

Enter Username: guest

Access Denied

3. Traffic Signal Indicator (If Elif Else)

Question:

Write a Python program that accepts a traffic signal color and displays the corresponding action.

- Red → Stop
- Yellow → Wait
- Green → Go
- Otherwise → Invalid Signal

Sample Output:

Enter Signal Color: Green

Go

4. Employee Bonus Calculator (If Elif Else)

Question:

Write a Python program to determine an employee's bonus percentage based on years of service.

- More than 10 years → 20% Bonus
- 5 to 10 years → 10% Bonus
- Less than 5 years → 5% Bonus

Sample Output:

Enter Years of Service: 8

Bonus: 10%

5. PIN Validation System (While Loop)

Question:

Write a Python program that keeps asking the user to enter a PIN until the correct PIN (1234) is entered.

Sample Output:

Enter PIN: 1111

Incorrect PIN

Enter PIN: 2222

Incorrect PIN

Enter PIN: 1234

Access Granted

6. Factory Machine Monitor (While Loop)

Question:

Write a Python program that keeps asking for machine temperature. Stop when the temperature exceeds 100°C and display a warning message.

Sample Output:

Enter Temperature: 75

Enter Temperature: 82

Enter Temperature: 95

Enter Temperature: 105

Warning! Machine Overheated.

7. Multiplication Table Generator (For Loop)

Question:

Write a Python program to display the multiplication table of a given number from 1 to 10.

Sample Output:

Enter Number: 7

7 x 1 = 7

7 x 2 = 14

7 x 3 = 21

...

7 x 10 = 70

8. Employee ID Generator (For Loop)

Question:

Write a Python program to generate employee IDs from EMP101 to EMP110.

Sample Output:

EMP101

EMP102

EMP103

EMP104

EMP105

EMP106

EMP107

EMP108

EMP109

EMP110

9. Hotel Room Matrix (Nested For Loop)

Question:

Write a Python program to display room numbers for a hotel with 4 floors and 5 rooms on each floor.

Sample Output:

101 102 103 104 105

201 202 203 204 205

301 302 303 304 305

401 402 403 404 405

10. Star Pattern for Event Banner (Nested For Loop)

Question:

Write a Python program to display a rectangular banner consisting of 6 rows and 8 columns of stars.

Sample Output:

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

11. Parking Fee Calculator (If Elif Else)

Question:

Write a Python program to calculate parking fees based on hours parked.

- Up to 2 hours → ₹20
- 3 to 5 hours → ₹50
- More than 5 hours → ₹100

Sample Output:

Enter Parking Hours: 4

Parking Fee: ₹50

12. Bus Seat Reservation Display (Nested For Loop)

Question:

Write a Python program to display seat numbers for 5 rows and 4 seats per row.

Sample Output:

```
R1S1 R1S2 R1S3 R1S4
R2S1 R2S2 R2S3 R2S4
R3S1 R3S2 R3S3 R3S4
R4S1 R4S2 R4S3 R4S4
R5S1 R5S2 R5S3 R5S4
```